CSA4704-DEEP LEARNING

NAME:HARITHI S **REG NO:192421151**

Experiment 12: CNN Architecture and Hyperparameter Comparison
Problem Statement: The recycling plant's CNN must run on a low-cost edge device. The engineer must balance accuracy against model size and training time. The same waste or CIFAR-10 dataset from Experiment 11. The objective is to compare CNN architectures and hyperparameters systematically. Use TensorFlow, Keras, NumPy, Matplotlib.
Tasks:
1. Vary the number of convolution layers (2, 3, 4).
2. Vary filter size (3×3, 5×5) and filter count.
3. Compare max pooling and average pooling, with and without batch normalisation.
4. Vary learning rate and batch size.
5. Tabulate accuracy, parameter count and training time for each configuration.

FORMULA:

1. Convolution Operation

$$\text{Output}(i, j) = \Sigma\square\ \Sigma\square\ X(i + m, j + n) \times K(m, n) + b$$

2. ReLU Activation Function

$$f(x) = \max(0, x)$$

3. Max Pooling

$$P(i, j) = \max\ \{\ X(i + m, j + n)\ \}$$

4. Average Pooling

$$P(i, j) = (1 / N)\ \Sigma\ X(i + m, j + n)$$

5. Batch Normalisation

$$\hat{x} = (x - \mu) / \sqrt{(\sigma^2 + \varepsilon)}$$

$$y = \gamma\hat{x} + \beta$$

6. Categorical Cross-Entropy Loss

$L = -\Sigma_i y_i \log(\hat{y}_i)$

7. Softmax Activation

$\hat{y}_i = e^{z_i} / \Sigma_j e^{z_j}$

8. Classification Accuracy

Accuracy = (Correct Predictions / Total Predictions) × 100%

9. Trainable Parameters in a Convolution Layer

Parameters = $(K \times K \times C_{in} \times C_{out}) + C_{out}$

10. Gradient Descent Update

$W(new) = W(old) - \eta \times \partial L/\partial W$

11. Training Time

Training Time = End Time − Start Time

CODE:

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import time
from tensorflow.keras import layers, models

# 1. Initialize
…
# 10. Save results
df.to_csv("experiment12_cnn_results.csv", index=False)

print("\nExperiment 12 completed successfully!")
print("Results saved as experiment12_cnn_results.csv")
```

OUTPUT:

Dataset loaded successfully!

Training images: (20000, 32, 32, 3)
Testing images: (5000, 32, 32, 3)

Training: Baseline: 2 layers
Accuracy: 29.22 %
Parameters: 5562
Training time: 20.58 seconds

Training: 3 convolution layers
Accuracy: 39.64 %
Parameters: 24570
Training time: 17.73 seconds

Training: 4 convolution layers
Accuracy: 50.78 %
Parameters: 99450
Training time: 19.11 seconds

Training: 5x5 filters
Accuracy: 22.46 %
Parameters: 14522
Training time: 15.34 seconds

Training: Average pooling
Accuracy: 38.68 %
Parameters: 5562
Training time: 16.21 seconds

Training: Without batch normalization
Accuracy: 33.34 %
Parameters: 5418
Training time: 12.33 seconds

Training: Learning rate 0.0001
Accuracy: 32.2 %
Parameters: 5562
Training time: 12.26 seconds

Training: Batch size 64
Accuracy: 27.74 %

Parameters: 5562
Training time: 11.9 seconds

===== FINAL COMPARISON TABLE =====

	Configuration	Conv Layers	Filter Size	Pooling	Batch Normalization	Learning Rate	Batch Size	Test Accuracy (%)	Parameters	Training Time (sec)
0	Baseline: 2 layers	2	3x3	max	True	0.0010	32	29.22	5562	20.58
1	3 convolution layers	3	3x3	max	True	0.0010	32	39.64	24570	17.73
2	4 convolution layers	4	3x3	max	True	0.0010	32	50.78	99450	19.11
3	5x5 filters	2	5x5	max	True	0.0010	32	22.46	14522	15.34
4	Average pooling	2	3x3	average	True	0.0010	32	38.68	5562	16.21
5	Without batch normalization	2	3x3	max	False	0.0010	32	33.34	5418	12.33
6	Learning rate 0.0001	2	3x3	max	True	0.0001	32	32.20	5562	12.26
7	Batch size 64	2	3x3	max	True	0.0010	64	27.74	5562	11.90

Highest accuracy:

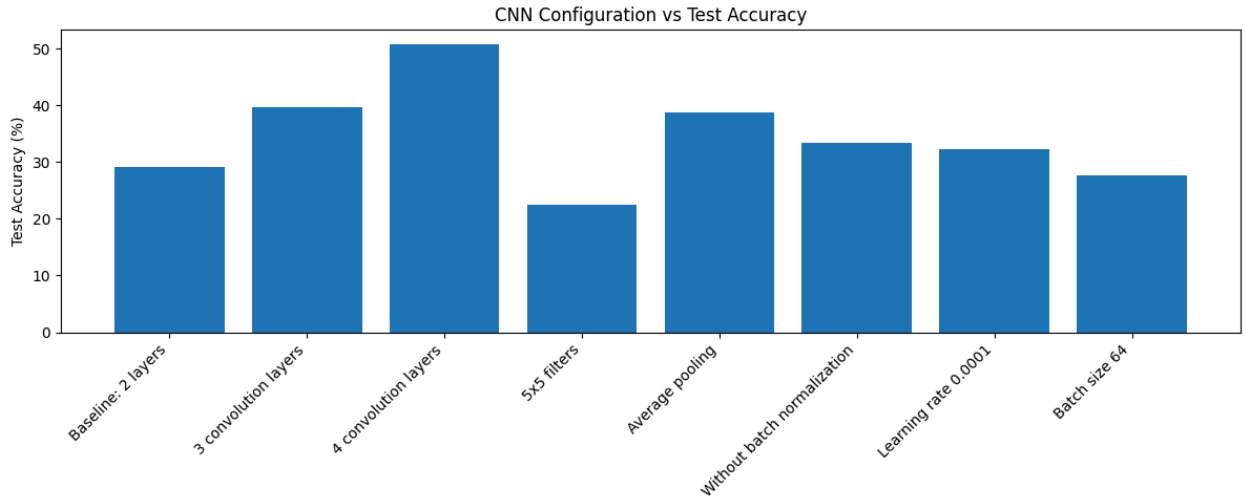
	Configuration	Conv Layers	Filter Size	Pooling	Batch Normalization	Learning Rate	Batch Size	Test Accuracy (%)	Parameters	Training Time (sec)
2	4 convolution layers	4	3x3	max	True	0.001	32	50.78	99450	19.11

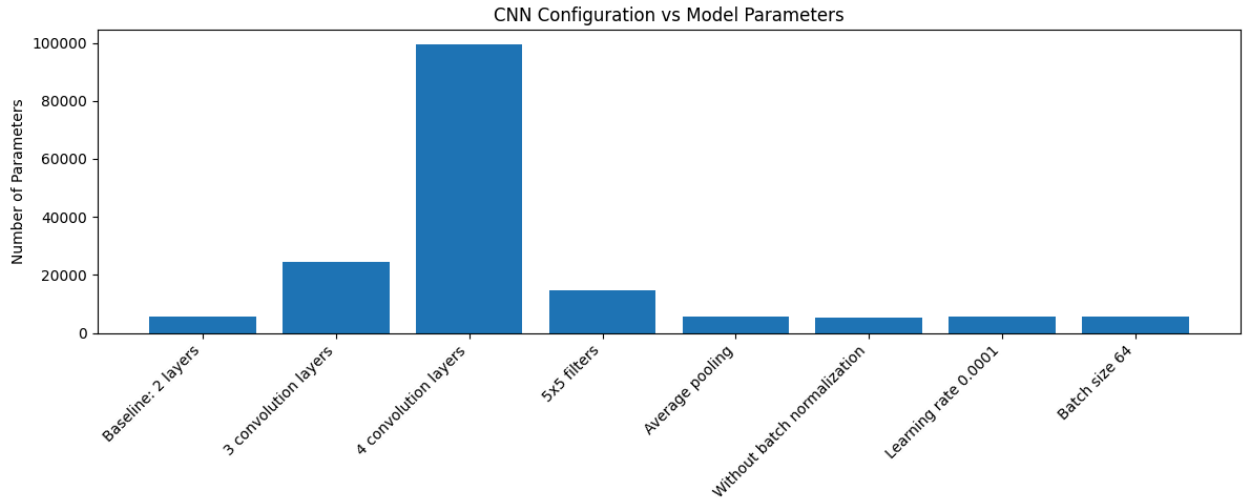
Smallest model:

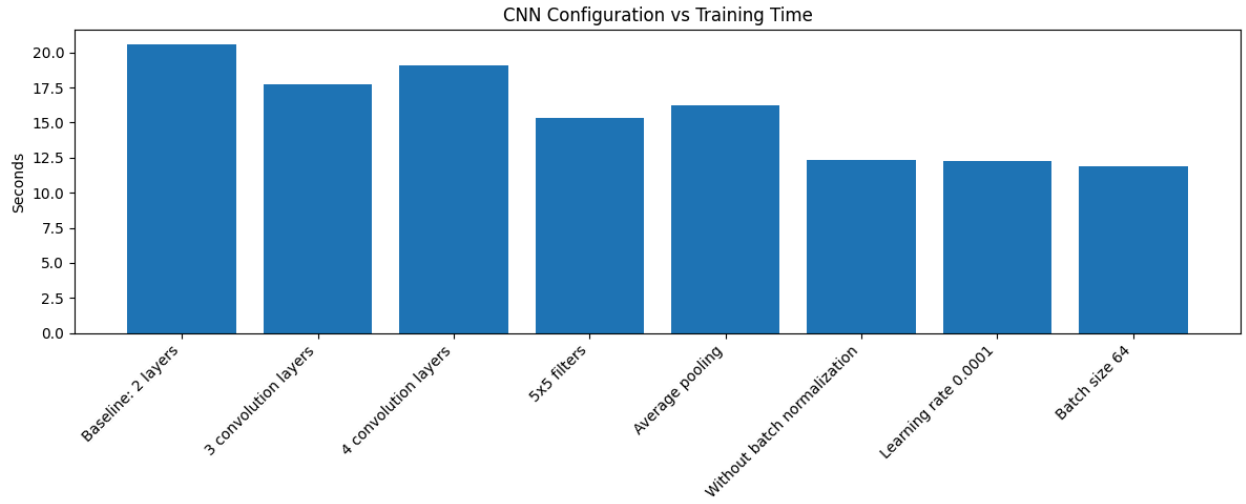
	Configuration	Conv Layers	Filter Size	Pooling	Batch Normalization	Learning Rate	Batch Size	Test Accuracy (%)	Parameters	Training Time (sec)
5	Without batch normalization	2	3x3	max	False	0.001	32	33.34	5418	12.33

Fastest training:

	Configuration	Conv Layers	Filter Size	Pooling	Batch Normalization	Learning Rate	Batch Size	Test Accuracy (%)	Parameters	Training Time (sec)
7	Batch size 64	2	3x3	max	True	0.001	64	27.74	5562	11.9







Experiment 12 completed successfully!
Results saved as experiment12_cnn_results.csv